

Note that $h_{p^k}^-$ is *easy* to compute for p^k of reasonable size, and we will give a formula for it in Corollary 10.5.27. On the other hand, it is much more difficult to compute $h_{p^k}^+$. For instance, the value of h_p^+ is known rigorously only for $p \leq 61$ (for which it is equal to 1), under the generalized Riemann hypothesis (GRH) for $p \leq 163$ (for which it is also equal to 1, except for $h_{163}^+ = 4$), although thanks in particular to work of R. Schoof, a conjectural value can be given for much larger p ; see pages 420–423 in [Was].

Finally, we give without proof the following fundamental result concerning the unit group and the class group.

Theorem 3.5.24. *Let $K = \mathbb{Q}(\zeta_{p^k})$ and let $U(K)$ be the unit group of K . The subgroup of cyclotomic units of $U(K)$ (see Definition 3.5.16) is a subgroup of finite index of $U(K)$, and this index is equal to $h_{p^k}^+$.*

3.6 Stickelberger's Theorem

3.6.1 Introduction and Algebraic Setting

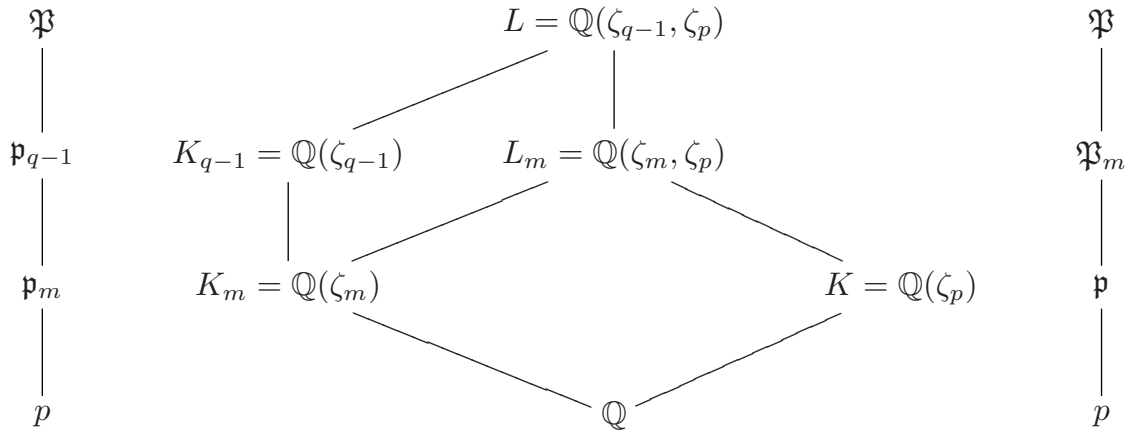
In this section, let K be a number field, for the moment arbitrary, and let $h(K)$ be its class number. One of the most important applications of the class group is that for any ideal $\mathfrak{a}$ of K the ideal $\mathfrak{a}^{h(K)}$ is a principal ideal. This is in fact the reason for which (in another language) it was introduced by Kummer in his work on Fermat's last theorem. Together with results on the unit group, which is inseparable from the class group, this allows us to treat K almost as if it had unique factorization (as foremost example, see the proof of FLT I for regular primes, i.e., Proposition 6.9.8).

In the case of a cyclotomic field (and more generally of an Abelian extension of $\mathbb{Q}$, which, by the Kronecker–Weber Theorem 3.5.13, is simply a subfield of a cyclotomic field), there is a more powerful theorem, due to Stickelberger, which gives more precise information about the Galois structure of the class group, hence also tells us how to construct principal ideals from arbitrary ideals. This theorem has a large number of important applications, of which we will see three in this book: the Eisenstein reciprocity law, the Hasse–Davenport relations, and Catalan's conjecture. Although perhaps less “basic” than the rest of this chapter, it is important to study it at this point, and since it is slightly less classical than class and unit groups (although it can be found in many excellent textbooks such as [Was]), we give completely detailed proofs. In fact, we will give *two* proofs. In this chapter we give the classical one. As the reader will see, it is quite lengthy and seems roundabout. It uses in a fundamental way the prime ideal decomposition of Gauss sums, and some additional technical machinery. In Chapter 11, however, we will see that thanks to the Gross–Koblitz formula, Gauss sums can be reinterpreted via Morita's p -adic gamma function $\Gamma_p(x)$. Thanks to this interpretation, in

Section 11.7.3 we will give a much shorter and natural proof of the Stickelberger congruence.

We begin by explaining the setting in which we work. Let p be a prime number, let $\zeta = \zeta_p$ be a primitive p th root of unity; set $K = \mathbb{Q}(\zeta)$, so that $\mathbb{Z}_K = \mathbb{Z}[\zeta]$; let $\pi = 1 - \zeta$, and let $\mathfrak{p} = \pi\mathbb{Z}_K$ be the unique ideal above p , hence such that $p\mathbb{Z}_K = \mathfrak{p}^{p-1}$. Finally, let $q = p^f$ be a power of p with $f \geq 1$, and let $L = \mathbb{Q}(\zeta_{q-1}, \zeta_p)$. Since p and $q - 1$ are coprime, the fields $\mathbb{Q}(\zeta_{q-1})$ and $K = \mathbb{Q}(\zeta_p)$ are linearly disjoint, and in fact it is clear that $L = \mathbb{Q}(\zeta_{p(q-1)})$, although we will never use this. Later we will also have an integer m dividing $q - 1$, and we will consider the fields $\mathbb{Q}(\zeta_m) \subset \mathbb{Q}(\zeta_{q-1})$ and $\mathbb{Q}(\zeta_m, \zeta_p) \subset L$. Since so many different cyclotomic fields are involved, we will use the following notation, which is probably the least confusing. For any integer n we will denote by K_n the cyclotomic field $\mathbb{Q}(\zeta_n)$ and for $m \mid (q - 1)$ we will denote by L_m the field $\mathbb{Q}(\zeta_m, \zeta_p)$. Thus for example $K = K_p$, $\mathbb{Q}(\zeta_{q-1}) = K_{q-1}$, and $L = L_{q-1}$.

Let $\mathfrak{P}$ be a *fixed* prime ideal of L above $\mathfrak{p}$. The prime ideal of L_m below $\mathfrak{P}$ will be denoted by $\mathfrak{P}_m$, so that $\mathfrak{P} = \mathfrak{P}_{q-1}$, and if $K_m \subset K_{q-1}$ the prime ideal below $\mathfrak{P}$ will be denoted by $\mathfrak{p}_m$. Since $p \nmid q - 1$, we have $e(\mathfrak{p}_{q-1}/p) = 1$; hence by Proposition 3.5.18, $f(\mathfrak{p}_{q-1}/p) = f$ since $q = p^f$ is evidently the smallest power of p congruent to 1 modulo $q - 1$. It follows that $\mathfrak{P}$ is unramified in the extension L/K and that we also have $f(\mathfrak{P}/\mathfrak{p}) = f$. The following diagram summarizes all of the above notation.



3.6.2 Instantiation of Gauss Sums

In Chapter 2 we introduced the Gauss sum $\tau(\chi, \psi)$ associated with a multiplicative and an additive character on a finite field $\mathbb{F}_q$, and we have seen for instance that $|\tau(\chi, \psi)|^2 = q$ when the characters are nontrivial. Since the values of ψ are in $K = \mathbb{Q}(\zeta)$ and the values of χ are in $K_{q-1} = \mathbb{Q}(\zeta_{q-1})$, it follows that $\tau(\chi, \psi) \in L$. In the present subsection we will see that we can give much more precise information. In particular, we will give the exact

description of the principal ideal generated by $\tau(\chi, \psi)$, and congruences satisfied by the Gauss sum. As mentioned above, in Section 11.7.3 we will see how to compute $\tau(\chi, \psi)$ in terms of the p -adic gamma function, leading to a simpler proof of the Stickelberger formula.

To be able to compute $\tau(\chi, \psi)$ we must *instantiate* all the mathematical objects that occur (otherwise, the computation would not make sense), in other words, $\mathbb{F}_q$ and the characters χ and ψ . To instantiate $\mathbb{F}_q$ we use the fixed prime ideal $\mathfrak{P}$ of L chosen above, and set $\mathbb{F}_q = \mathbb{Z}_L/\mathfrak{P}$, which is an extension of $\mathbb{F}_p = \mathbb{Z}_K/\mathfrak{p}$. By Proposition 2.5.4, any additive character ψ has the form $\psi = \psi_b$ for a unique $b \in \mathbb{F}_q$, where $\psi_b(x) = \zeta^{\text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(bx)}$. Since we will almost always use $\psi = \psi_1$, we will simply write $\tau(\chi)$ instead of $\tau(\chi, \psi_1)$, although it will occasionally be useful to use ψ_b because of the formula $\tau(\chi, \psi_b) = \chi(b)^{-1} \tau(\chi, \psi_1)$ of Lemma 2.5.6. To instantiate multiplicative characters, we need a lemma.

Lemma 3.6.1. *Let $\mu_{q-1} = \{\zeta_{q-1}^j, 0 \leq j \leq q-2\} \subset L$ be the group of $(q-1)$ st roots of unity in L . The map $u_{\mathfrak{P}}$ sending $x \in \mu_{q-1}$ to its class modulo $\mathfrak{P}$ in $\mathbb{F}_q^* = (\mathbb{Z}_L/\mathfrak{P})^*$ is a group isomorphism.*

Proof. It is clear that $u_{\mathfrak{P}}$ is a group homomorphism, and since μ_{q-1} and $\mathbb{F}_q^*$ both have $q-1$ elements it is sufficient to show that the kernel of $u_{\mathfrak{P}}$ is equal to 1. Now

$$\prod_{1 \leq k \leq q-2} (x - \zeta_{q-1}^k) = \frac{x^{q-1} - 1}{x - 1};$$

hence $\prod_{1 \leq k \leq q-2} (1 - \zeta_{q-1}^k) = q-1$. However, $u_{\mathfrak{P}}(\zeta_{q-1}^k) = 1$ means that $\mathfrak{P} \mid (\zeta_{q-1}^k - 1)$, hence if $1 \leq k \leq q-2$ this implies that $\mathfrak{P} \mid (q-1)$, which is absurd since $\mathfrak{P} \mid p$ and p is coprime to $q-1 = p^f - 1$. $\square$

Definition 3.6.2. *We define $\omega_{\mathfrak{P}} = u_{\mathfrak{P}}^{-1}$, in other words the unique group isomorphism from $\mathbb{F}_q^* = (\mathbb{Z}_L/\mathfrak{P})^*$ to μ_{q-1} such that $\omega_{\mathfrak{P}}(x) \equiv x \pmod{\mathfrak{P}}$.*

By definition $\omega_{\mathfrak{P}}$ is a multiplicative character of $\mathbb{F}_q$ of order equal to $q-1$ since it is an isomorphism (in the p -adic context it will be called the Teichmüller character; see Proposition 4.3.4). Since $\mathbb{F}_q^*$ is a cyclic group of order $q-1$, its group of characters is also cyclic of order $q-1$, so it follows that any multiplicative character χ on $\mathbb{F}_q$ has the form $\omega_{\mathfrak{P}}^{-r}$ for a unique r such that $0 \leq r < q-1$ (equivalently, by transitivity of the Galois action on prime ideals above $\mathfrak{p}$, any multiplicative character χ on $\mathbb{F}_q$ is one of the $\omega_{\mathfrak{Q}}$ for a suitable prime ideal $\mathfrak{Q}$ above p). Thus with an evident abuse of notation, any Gauss sum over $\mathbb{F}_q$ has the form $\tau(\omega_{\mathfrak{P}}^{-r}, \psi_b)$ for a unique $r \in \mathbb{Z}/(q-1)\mathbb{Z}$ and a unique $b \in \mathbb{F}_q$. This now depends only on the triple $(\mathfrak{P}, r, b)$; hence we will compute its properties only in terms of this triple. In fact, since $\tau(\omega_{\mathfrak{P}}^{-r}, \psi_b) = \omega_{\mathfrak{P}}^r(b) \tau(\omega_{\mathfrak{P}}^{-r})$, we may assume that $b = 1$. Finally, since in this section $\mathfrak{P}$ is fixed, for notational convenience we will simply write ω instead

of $\omega_{\mathfrak{P}}$, but we have to keep in mind for later that the definition of ω depends on the choice of the ideal $\mathfrak{P}$ of L above $\mathfrak{p}$. In fact, since by Galois theory all other ideals have the form $\sigma(\mathfrak{P})$ for $\sigma \in \text{Gal}(L/K)$, the following lemma makes explicit the dependence on $\mathfrak{P}$:

Lemma 3.6.3. (1) *For all $\sigma \in \text{Gal}(L/\mathbb{Q})$ we have*

$$\omega_{\sigma(\mathfrak{P})} = \sigma \circ \omega_{\mathfrak{P}} \circ \sigma^{-1}.$$

(2) *For all $\sigma \in \text{Gal}(L/\mathbb{Q})$ and $r \in \mathbb{Z}$ we have*

$$\tau(\omega_{\sigma(\mathfrak{P})}^{-r}) = \sigma(\tau(\omega_{\mathfrak{P}}^{-r})).$$

(3) *If $m \mid (q-1)$ and $x \in K_m$ then $\omega_{\mathfrak{P}}^{-(q-1)/m}(x)$ depends only on the ideal $\mathfrak{p}_m$ of K_m below $\mathfrak{P}$ and not on $\mathfrak{P}$ itself.*

(4) *If $m \mid (q-1)$ then $\tau(\omega_{\mathfrak{P}}^{-(q-1)/m})^m \in K_m$, and it depends only on the ideal $\mathfrak{p}_m$ of K_m below $\mathfrak{P}$ and not on $\mathfrak{P}$ itself.*

Proof. Since $\omega_{\mathfrak{P}}(\sigma^{-1}(x))$ is a $(q-1)$ st root of unity, so is $\sigma(\omega_{\mathfrak{P}}(\sigma^{-1}(x)))$. Furthermore, $\omega_{\mathfrak{P}}(\sigma^{-1}(x)) \equiv \sigma^{-1}(x) \pmod{\mathfrak{P}}$, hence applying σ we have $\sigma(\omega_{\mathfrak{P}}(\sigma^{-1}(x))) \equiv x \pmod{\sigma(\mathfrak{P})}$. Since these two properties characterize $\omega_{\sigma(\mathfrak{P})}$, (1) follows. Set $\mathfrak{P}_1 = \sigma(\mathfrak{P})$. With evident notation we have

$$\tau(\omega_{\mathfrak{P}_1}^{-r}) = \sum_{x \in (\mathbb{Z}_L/\mathfrak{P}_1)^*} \omega_{\mathfrak{P}_1}(x)^{-r} \zeta_p^{\text{Tr}(\mathbb{Z}_L/\mathfrak{P}_1)/(\mathbb{Z}/p\mathbb{Z})(x)},$$

so that setting $x = \sigma(y)$ and using (1) we obtain

$$\sigma^{-1}(\tau(\omega_{\mathfrak{P}_1}^{-r})) = \sum_{y \in (\mathbb{Z}_L/\mathfrak{P})^*} \omega_{\mathfrak{P}}(y)^{-r} \zeta_p^{\text{Tr}(\mathbb{Z}_L/\mathfrak{P})/(\mathbb{Z}/p\mathbb{Z})(y)} = \tau(\omega_{\mathfrak{P}}^{-r}),$$

proving (2). For (3) and (4), set $d = (q-1)/m$, a notation that we shall in fact use in the rest of this chapter. By definition, $\omega_{\mathfrak{P}}^d(x) \equiv x^d \pmod{\mathfrak{P}}$, and $\omega_{\mathfrak{P}}^d(x)$ is an m th root of unity, and in particular belongs to K_m . If in addition $x \in K_m$, then $\omega_{\mathfrak{P}}^d(x) \equiv x^d \pmod{\mathfrak{P} \cap K_m}$, proving (3) since $\omega_{\mathfrak{P}}^d(x)$ is characterized by the above two properties. For (4), Corollary 2.5.15 tells us that $\tau(\omega_{\mathfrak{P}}^{-d})^m \in K_m$. If $\mathfrak{P}_1$ is some other prime ideal of L above the same prime ideal $\mathfrak{p}_m$ of K_m , by transitivity of the Galois action on prime ideals there exists $\sigma \in \text{Gal}(L/K_m)$ such that $\mathfrak{P}_1 = \sigma(\mathfrak{P})$. Thus by (2) we have $\tau(\omega_{\mathfrak{P}_1}^{-d})^m = \sigma(\tau(\omega_{\mathfrak{P}}^{-d})^m)$. Since σ leaves K_m fixed it does not act on the values of the multiplicative characters, but only on the additive characters; hence by Lemma 2.5.6 we have for some integer a coprime to p ,

$$\sigma(\tau(\omega_{\mathfrak{P}_1}^{-d})^m) = \tau(\omega_{\mathfrak{P}}^{-d}, \psi_a)^m = (\omega_{\mathfrak{P}}(a)^d \tau(\omega_{\mathfrak{P}}^{-d}))^m = \tau(\omega_{\mathfrak{P}}^{-d})^m,$$

since $\omega_{\mathfrak{P}}(a)$ is a $(q-1)$ st root of unity, proving (4). $\square$

3.6.3 Prime Ideal Decomposition of Gauss Sums

We begin by proving the following congruence for Jacobi sums.

Proposition 3.6.4. *If a and b are integers such that $1 \leq a, b \leq q-2$ we have $J(\omega^{-a}, \omega^{-b}) \equiv 0 \pmod{\mathfrak{P}}$ if $a+b \geq q$, and otherwise*

$$J(\omega^{-a}, \omega^{-b}) \equiv -\binom{a+b}{a} \equiv -\frac{(a+b)!}{a!b!} \pmod{\mathfrak{P}}.$$

Proof. For any $x \in \mathbb{F}_q^* = (\mathbb{Z}_L/\mathfrak{P})^*$ we let $x' \in \mu_{q-1} \subset \mathbb{Z}_L$ be such that $x' \equiv x \pmod{\mathfrak{P}}$, so that $\omega(x) = x'$. Setting $c = q-1-b$ and noting that $\omega^{q-1} = \varepsilon$ we see that $\omega^{-b} = \omega^c$. In particular, for $x \neq 1$ we have $\omega^{-b}(1-x) = \omega^c(1-x) \equiv (1-x')^c \pmod{\mathfrak{P}}$, and we note that this congruence is trivially also true for $x = 1$ since $b < q-1$, and hence $c > 0$. Thus

$$\begin{aligned} J(\omega^{-a}, \omega^{-b}) &= \sum_{x \in \mathbb{F}_q^*} \omega^{-a}(x) \omega^{-b}(1-x) \equiv \sum_{x \in \mathbb{F}_q^*} x'^{-a} (1-x')^c \\ &\equiv \sum_{x \in \mathbb{F}_q^*} x^{-a} \sum_{0 \leq j \leq b} \binom{c}{j} (-1)^j x^j \\ &\equiv \sum_{0 \leq j \leq c} (-1)^j \binom{c}{j} \sum_{x \in \mathbb{F}_q^*} x^{j-a} \pmod{\mathfrak{P}}. \end{aligned}$$

Thanks to Lemma 2.5.1 we see that the inner sum vanishes except if $j \equiv a \pmod{q-1}$, in other words if $j = a$ since $1 \leq a \leq q-2$. Thus if $a > c$, in other words $a+b > q-1$, all the inner sums vanish, so we get $J(\omega^{-a}, \omega^{-b}) \equiv 0 \pmod{\mathfrak{P}}$. On the other hand, if $a \leq c$ we have a single nonzero inner sum for $j = a$, which is congruent to -1 modulo p , so that

$$J(\omega^{-a}, \omega^{-b}) \equiv (-1)^{a-1} \binom{c}{a} \pmod{\mathfrak{P}},$$

and the proposition follows since

$$\begin{aligned} \binom{c}{a} &= \frac{(q-1-b)(q-2-b) \cdots (q-a-b)}{a!} \\ &\equiv (-1)^a \frac{(b+1)(b+2) \cdots (b+a)}{a!} \equiv (-1)^a \binom{a+b}{a} \pmod{p}. \end{aligned}$$

□

Remarks. (1) It is easy to show that when $a+b \geq q$ we have $\binom{a+b}{a} \equiv 0 \pmod{p}$ (see Exercise 25), so the second congruence of the proposition is in fact valid in all generality.

(2) This proposition again shows the analogy between Jacobi sums and the beta function.

Definition 3.6.5. Let r be an integer such that $0 \leq r < q - 1$, and let $r = \sum_{0 \leq i < f} r_i p^i$ be the decomposition of r in base p with $0 \leq r_i \leq p - 1$. We set $s_p(r) = \sum_{0 \leq i < f} r_i$ and $t_p(r) = \prod_{0 \leq i < f} (r_i)!$. If $r \in \mathbb{Z}$ is arbitrary, we define the $(q - 1)$ -periodic functions $s(r)$ and $t(r)$ by setting $s(r) = s_p(r \bmod q - 1)$ and $t(r) = t_p(r \bmod q - 1)$, where $r \bmod q - 1$ is the unique integer in $[0, q - 1[$ congruent to r modulo $q - 1$.

Stickelberger's basic result is the following.

Theorem 3.6.6. For all $r \in \mathbb{Z}$ we have

$$\frac{\tau(\omega^{-r})}{(\zeta - 1)^{s(r)}} \equiv -\frac{1}{t(r)} \pmod{\mathfrak{P}}.$$

Proof. By periodicity we may assume that $0 \leq r < q - 1$. We prove the theorem by induction on $s(r) = s_p(r)$. If $s(r) = 0$ we have $r = 0$, hence $t(0) = 1$ and $\tau(\omega^0) = \tau(\varepsilon) = -1$ by Lemma 2.5.8. The crucial case to be proved is the case $s(r) = 1$, in other words, $r = p^k$. In that case $t(r) = 1$, and since by Lemma 2.5.8 (3) we have $\tau(\omega^{-p^a}) = \tau(\omega^{-a})$, it follows that we may assume that $r = 1$. Since ω is a nontrivial character we have

$$\tau(\omega^{-1}) = \sum_{x \in \mathbb{F}_q^*} \omega^{-1}(x) \left(\zeta_p^{\text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(x)} - 1 \right).$$

Since $\mathfrak{p} = (\zeta_p - 1)\mathbb{Z}_K$ we have

$$\frac{\zeta_p^m - 1}{\zeta_p - 1} = \sum_{0 \leq j < m} \zeta_p^j \equiv \sum_{0 \leq j < m} 1 \equiv m \pmod{\mathfrak{p}},$$

hence

$$\frac{\tau(\omega^{-1})}{\zeta_p - 1} \equiv \sum_{x \in \mathbb{F}_q^*} \omega^{-1}(x) \text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(x) \pmod{\mathfrak{p}}.$$

Now $\text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(x) = \sum_{0 \leq i < f} x^{p^i} \in \mathbb{F}_p$, and on the other hand, by definition, $\omega^{-1}(x) \equiv x^{-1} \pmod{\mathfrak{P}}$. It follows that

$$\frac{\tau(\omega^{-1})}{\zeta_p - 1} \equiv \sum_{0 \leq i < f} \sum_{x \in \mathbb{F}_q^*} x^{p^i - 1} \pmod{\mathfrak{P}}.$$

Now again by Lemma 2.5.1 the inner sum vanishes if $1 \leq i < f$, and it is congruent to -1 modulo p if $i = 0$. It follows that $\tau(\omega^{-1})/(\zeta_p - 1) \equiv -1 \pmod{\mathfrak{P}}$, proving the theorem when $s(r) = 1$.

Now let r be such that $0 \leq r < q - 1$ with $s(r) > 1$, assume by induction that the theorem is true for all $r' < q - 1$ such that $s(r') < s(r)$, and let $r = \sum_{0 \leq i < f} r_i p^i$ with $0 \leq r_i \leq p - 1$. Thanks once again to Lemma 2.5.8 (3) we may assume that $r_0 \geq 1$. It follows in particular that $s(r - 1) = s(r) - 1$

and $r - 1 \geq 1$. Since all the characters involved are nontrivial, by Corollary 2.5.17 we have

$$\tau(\omega^{-1})\tau(\omega^{-(r-1)}) = J(\omega^{-1}, \omega^{-(r-1)})\tau(\omega^{-r}) .$$

Using Proposition 3.6.4 we know that

$$J(\omega^{-1}, \omega^{-(r-1)}) \equiv -\binom{r}{1} \equiv -r \equiv -r_0 \pmod{\mathfrak{P}} .$$

Since $1 \leq r_0 \leq p-1$, r_0 is invertible modulo $\mathfrak{P}$, so by induction and the case $r = 1$ we see that

$$\begin{aligned} \frac{\tau(\omega^{-r})}{(\zeta_p - 1)^{s(r)}} &\equiv -\frac{1}{r_0} \frac{\tau(\omega^{-1})}{\zeta_p - 1} \frac{\tau(\omega^{-(r-1)})}{(\zeta_p - 1)^{s(r-1)}} \\ &\equiv -\frac{1}{r_0} \frac{1}{t(r-1)} \equiv -\frac{1}{t(r)} \pmod{\mathfrak{P}} , \end{aligned}$$

since $t(r) = r_0 t(r-1)$ when $r_0 \neq 0$, proving our induction hypothesis and hence the theorem. $\square$

To use this theorem, we will also need the following lemma.

Lemma 3.6.7. *As usual denote by $\{z\}$ the fractional part of a real number z .*

(1) *For all $r \in \mathbb{Z}$ we have*

$$s(r) = (p-1) \sum_{0 \leq i < f} \left\{ \frac{p^i r}{q-1} \right\} .$$

(2) *For $0 \leq r < q-1$ we have*

$$t(r) \equiv (-p)^{-v_p(r!)} r! \pmod{p} .$$

Proof. (1). Both sides of the formula are periodic of period dividing $q-1$; hence we may assume that $0 \leq r < q-1$, so that $r = \sum_{0 \leq j < f} r_j p^j$ with $0 \leq r_j \leq p-1$. For $0 \leq i \leq f-1$ we have

$$\begin{aligned} p^i r &= \sum_{0 \leq j < f-i-1} r_j p^{j+i} + \sum_{f-i \leq j < f} r_j p^{j+i} \\ &\equiv \sum_{0 \leq j < f-i-1} r_j p^{j+i} + \sum_{f-i \leq j < f} r_j p^{j+i-f} \pmod{q-1} , \end{aligned}$$

hence

$$\left\{ \frac{p^i r}{q-1} \right\} = \frac{1}{q-1} \left(\sum_{0 \leq j < f-i-1} r_j p^{j+i} + \sum_{f-i \leq j < f} r_j p^{j+i-f} \right) .$$

It follows that

$$\sum_{0 \leq i < f} \left\{ \frac{p^i r}{q-1} \right\} = \frac{1}{q-1} \sum_{0 \leq j < f} r_j S_j ,$$

where

$$S_j = \sum_{0 \leq i < f-j-1} p^{j+i} + \sum_{f-j \leq i < f} p^{j+i-f} = \sum_{0 \leq i < f} p^i = \frac{p^f - 1}{p - 1} = \frac{q - 1}{p - 1} ,$$

proving (1).

(2) is easily proved by induction on r : it is trivially true for $r \leq 1$. Assume $r \geq 2$ and that the formula is true for $r - 1$, and let $r = \sum_{k \leq j \leq f-1} r_j p^j$ be the base- p decomposition of r , with $0 \leq r_j \leq p - 1$ and $r_k \neq 0$. Since

$$r - 1 = \sum_{0 \leq j \leq k-1} (p - 1)p^j + (r_k - 1)p^k + \sum_{k+1 \leq j \leq f-1} r_j p^j$$

it follows from Wilson's theorem that

$$t(r - 1) \equiv (-1)^k (r_k - 1)! \prod_{k+1 \leq j \leq f-1} (r_j)! \pmod{p} ,$$

hence that

$$t(r) \equiv (-1)^k r_k t(r - 1) \equiv \frac{r}{(-p)^{v_p(r)}} t(r - 1) \pmod{p} .$$

The result follows by induction. $\square$

Corollary 3.6.8. *We have $v_{\mathfrak{P}}(\tau(\omega^{-r})) = s(r)$.*

Proof. By definition $t(r)$ is coprime to p hence is invertible modulo $\mathfrak{P}$, so by the theorem $v_{\mathfrak{P}}(\tau(\omega^{-r})) = s(r)v_{\mathfrak{P}}(\zeta - 1) = s(r)e(\mathfrak{P}/\mathfrak{p}) = s(r)$. $\square$

Definition 3.6.9. *Let $m \mid (q - 1)$ be such that $m > 1$.*

- (1) *For notational simplicity we will set $d = (q - 1)/m$.*
- (2) *If t is coprime to m we denote by σ_t the element of $\text{Gal}(L_m/K)$ such that $\sigma_t(\zeta_m) = \zeta_m^t$ (and of course leaving fixed ζ_p).*

Although strictly speaking σ_t depends on m , since these maps are compatible under restriction, there is no possibility of confusion.

Proposition 3.6.10. *Let $m \mid (q - 1)$, set $d = (q - 1)/m$, and recall that $L_m = \mathbb{Q}(\zeta_m, \zeta_p)$ and that $\mathfrak{P}_m$ is the prime ideal of L_m below $\mathfrak{P}$. Then*

$$\tau(\omega^{-rd})\mathbb{Z}_{L_m} = \prod_{t \in (\mathbb{Z}/m\mathbb{Z})^* / \langle p \rangle} \sigma_t^{-1}(\mathfrak{P}_m)^{s(rtd)} .$$

Proof. First note that the values of ω^{-rd} are in K_m , so that $\tau(\omega^{-rd}) \in L_m$. Since $\mathfrak{P}_m$ is a prime ideal of L_m above $\mathfrak{p}$, by Galois theory all the prime ideals of L_m above $\mathfrak{p}$ have the form $\sigma(\mathfrak{P}_m)$ for $\sigma \in \text{Gal}(L_m/K) \simeq \text{Gal}(K_m/\mathbb{Q}) \simeq (\mathbb{Z}/m\mathbb{Z})^*$. By definition of the Gauss sum we have $\sigma_t(\tau(\omega^{-rd})) = \tau(\omega^{-rtd})$. Thus by the above corollary

$$v_{\sigma_t^{-1}(\mathfrak{P}_m)}(\tau(\omega^{-rd})) = v_{\mathfrak{P}_m}(\sigma_t(\tau(\omega^{-rd}))) = v_{\mathfrak{P}_m}(\tau(\omega^{-rtd})) = s(rtd) ,$$

since $\mathfrak{P}$ is unramified over $\mathfrak{P}_m$. Furthermore, the decomposition group $D(\mathfrak{P}_m/\mathfrak{p})$ is isomorphic to $D(\mathfrak{p}_m/p)$; hence by Proposition 3.5.18 it is isomorphic to the cyclic subgroup of $\text{Gal}(L_m/K)$ generated by σ_p , using the same notation as above. This means that the ideals of L_m above $\mathfrak{p}$ are obtained once and only once as $\sigma_t^{-1}(\mathfrak{P}_m)$ for $\sigma_t \in \text{Gal}(L_m/K)/D(\mathfrak{P}_m/\mathfrak{p})$, in other words for $t \in (\mathbb{Z}/m\mathbb{Z})^*/\langle p \rangle$. Finally, since $\tau(\omega^{-rd})\tau(\omega^{-rd}) = q = p^f$ for $r \neq 0$, it follows that the only ideals of L_m dividing $\tau(\omega^{-rd})$ are the ideals above p , hence the ideals above $\mathfrak{p}$, in other words the $\sigma_t^{-1}(\mathfrak{P}_m)$ for $t \in (\mathbb{Z}/m\mathbb{Z})^*/\langle p \rangle$, so the proposition follows. $\square$

Corollary 3.6.11. *Let $m \mid (q-1)$, and recall that $K_m = \mathbb{Q}(\zeta_m)$ and that $\mathfrak{p}_m$ is the prime ideal of K_m below $\mathfrak{P}$. Then*

$$\tau(\omega^{-d})^m \mathbb{Z}_{K_m} = \prod_{t \in (\mathbb{Z}/m\mathbb{Z})^*/\langle p \rangle} \sigma_t^{-1}(\mathfrak{p}_m)^{v_t} ,$$

where

$$v_t = \frac{m}{p-1} s \left(t \frac{q-1}{m} \right) = m \sum_{0 \leq i < f} \left\{ \frac{p^i t}{m} \right\} .$$

Proof. By Corollary 2.5.15 we know that $\tau(\omega^{-d})^m \in K_m$. Since $e(\mathfrak{P}_m/\mathfrak{p}_m) = e(\mathfrak{p}/p) = p-1$ we have

$$v_{\mathfrak{p}_m}(\tau(\omega^{-d})^m) = (m/(p-1))v_{\mathfrak{P}_m}(\tau(\omega^{-d})) ,$$

so the corollary immediately follows from the proposition, the second formula for v_t coming from Lemma 3.6.7. $\square$

The above results are expressed much more nicely in the language of group rings, which we recall for the convenience of the reader.

Definition 3.6.12. *Let A be a commutative ring and let G be a finite abelian group. We define the group algebra $A[G]$ by*

$$A[G] = \left\{ \sum_{g \in G} a_g g, \ a_g \in A \right\} ,$$

in other words the set of formal linear combinations with coefficients in A of the elements of G , the addition law coming from that of A and the multiplication law from that of A and of the group law of G , in other words,

$$\left(\sum_{g \in G} a_g g \right) \left(\sum_{h \in G} b_h h \right) = \sum_{g, h \in G} a_g b_h gh = \sum_{g \in G} \left(\sum_{h \in G} a_h b_{h^{-1}g} \right) g.$$

It is immediately checked that $A[G]$ is a commutative ring. If E is a group (written multiplicatively, say) on which the group G acts, then E has a natural structure of a (left) $\mathbb{Z}[G]$ -module through the formula

$$\left(\sum_{g \in G} a_g g \right) \cdot v = \prod_{g \in G} (g \cdot v)^{a_g}.$$

If $\theta = \sum_{g \in G} a_g g$, it is customary to write v^θ instead of $\theta \cdot v$, so that the formula $\theta_1 \cdot (\theta_2 \cdot v) = (\theta_1 \theta_2) \cdot v$ translates into the identity $v^{\theta_1 \theta_2} = (v^{\theta_1})^{\theta_2}$, where it is essential that G be an *abelian* group.

The situation considered in practice will be the following. We will be given an Abelian extension $K/\mathbb{Q}$ with commutative Galois group G . The group G acts naturally on all natural objects linked to K , for instance on elements, units, ideals, ideal classes, etc., so by the above we have an action of $\mathbb{Z}[G]$, written in an exponential manner.

As a first application of this notation we have the following proposition, which is a restatement of Corollary 3.6.11.

Proposition 3.6.13. *Set*

$$\Theta = \sum_{t \in (\mathbb{Z}/m\mathbb{Z})^*} \left\{ \frac{t}{m} \right\} \sigma_t^{-1} = \frac{1}{m} \sum_{\substack{1 \leq t \leq m-1 \\ \gcd(t, m) = 1}} t \sigma_t^{-1}.$$

With the same notation as above we have $\tau(\omega^{-d})^m \mathbb{Z}_{K_m} = \mathfrak{p}_m^{m\Theta}$.

Proof. Let T be a system of representatives of $(\mathbb{Z}/m\mathbb{Z})^*$ modulo the cyclic subgroup $\langle p \rangle$ generated by p . Proposition 3.6.11 can be restated by saying that $\tau(\omega^{-d})^m \mathbb{Z}_{K_m} = \mathfrak{p}_m^{m\theta}$ with

$$\theta = \sum_{t \in T} \sum_{0 \leq i < f} \{p^i t/m\} \sigma_t^{-1}$$

(note that it would not make sense to replace T by $(\mathbb{Z}/m\mathbb{Z})^*/\langle p \rangle$ in the above expression since σ_t would not be defined). By definition, as t ranges in T and i ranges from 0 to $f-1$ the elements $p^i t$ modulo m range through $(\mathbb{Z}/m\mathbb{Z})^*$, so that

$$\theta = \sum_{t \in (\mathbb{Z}/m\mathbb{Z})^*} \{t/m\} \sigma_{t_1}^{-1},$$

where t_1 is the representative in T of the class of t modulo $\langle p \rangle$. Since the decomposition group of $\mathfrak{p}_m/p$ is the group generated by σ_p , it follows that $\mathfrak{p}_m^{m\theta} = \mathfrak{p}_m^{m\Theta}$, where Θ is as in the proposition. $\square$

The following corollary is one of the most important consequences.

Corollary 3.6.14. *Let $\mathbb{Q}(\zeta_m)$ be a cyclotomic field and let Θ be defined as above. For all fractional ideals $\mathfrak{a}$ of $\mathbb{Q}(\zeta_m)$ the ideal $\mathfrak{a}^{m\Theta}$ is a principal ideal.*

Proof. We know that in any ideal class there exists an integral ideal coprime to any fixed ideal, in particular to m . In other words, there exists an element α such that $\mathfrak{b} = \alpha\mathfrak{a}$ is an integral ideal coprime to m . If $\mathfrak{p}_m$ is a prime ideal dividing $\mathfrak{b}$ above some prime p we have $p \nmid m$; hence the proposition implies that $\mathfrak{p}_m^{m\Theta}$ is a principal ideal. Since this is true for every prime ideal dividing $\mathfrak{b}$, by multiplicativity it is true for $\mathfrak{b}$ itself, and hence for $\mathfrak{a}$. $\square$

3.6.4 The Stickelberger Ideal

Although the above results and in particular Corollary 3.6.14 are remarkable results, they suffer from the presence of m in the exponent. We are now going to show how to get rid of this exponent m . We keep all the above notation, and we set $G = \text{Gal}(K_m/\mathbb{Q})$, which is canonically isomorphic to $(\mathbb{Z}/m\mathbb{Z})^*$.

Definition 3.6.15. (1) We define the Stickelberger ideal $I_s(m)$ by $I_s(m) = \mathbb{Z}[G] \cap \Theta\mathbb{Z}[G]$.
 (2) If b is an integer coprime to m we define $\Theta_b \in \mathbb{Q}[G]$ by the formula $\Theta_b = (\sigma_b - b)\Theta$.

It is clear that $I_s(m)$ is an ideal of the commutative ring $\mathbb{Z}[G]$.

Lemma 3.6.16. *We have $\Theta_b \in I_s(m)$. More precisely, we have*

$$\Theta_b = - \sum_{1 \leq t \leq m-1, \gcd(t,m)=1} \left\lfloor \frac{bt}{m} \right\rfloor \sigma_t^{-1}.$$

Proof. Since evidently $\Theta_b \in \mathbb{Z}[G]\Theta$, we must show that $\Theta_b \in \mathbb{Z}[G]$. Setting $u = b^{-1}t$ we have

$$\sigma_b \Theta = \sum_{t \in (\mathbb{Z}/m\mathbb{Z})^*} \{t/m\} \sigma_{b^{-1}t}^{-1} = \sum_{u \in (\mathbb{Z}/m\mathbb{Z})^*} \{bu/m\} \sigma_u^{-1},$$

hence $\Theta_b = \sum_{u \in (\mathbb{Z}/m\mathbb{Z})^*} c_u \sigma_u^{-1}$ with

$$c_u = \{bu/m\} - b\{u/m\} \equiv bu/m - b(u/m) \equiv 0 \pmod{\mathbb{Z}},$$

proving the first part of the lemma. The formula follows by summing over $u \in [1, m-1]$ coprime to m , since in that case $\{u/m\} = u/m$. $\square$

Example. If $m = p$ is an odd prime and $b = 2$ we have

$$\Theta_2 = - \sum_{(p+1)/2 \leq t \leq p-1} \sigma_t^{-1}.$$

Lemma 3.6.17. *The ideal $I_s(m)$ is generated by the Θ_b as a $\mathbb{Z}$ -module (hence also as an ideal). More precisely, it is generated over $\mathbb{Z}$ by Θ_{m+1} and the Θ_b for $1 \leq b \leq m$ with $\gcd(b, m) = 1$.*

Proof. By definition an element $\delta \in I_s(m)$ has the form $\delta = \gamma\Theta$, where $\gamma\Theta \in \mathbb{Z}[G]$ and $\gamma \in \mathbb{Z}[G]$. If we write $\gamma = \sum_{t \in (\mathbb{Z}/m\mathbb{Z})^*} c_t \sigma_t$ with $c_t \in \mathbb{Z}$ we have

$$\gamma\Theta = \sum_{t, u \in (\mathbb{Z}/m\mathbb{Z})^*} c_t \{u/m\} \sigma_{t^{-1}u}^{-1} = \sum_{v \in (\mathbb{Z}/m\mathbb{Z})^*} d_v \sigma_v^{-1}$$

with

$$d_v = \sum_{t \in (\mathbb{Z}/m\mathbb{Z})^*} c_t \{tv/m\}.$$

Since $\gamma\Theta \in \mathbb{Z}[G]$ we have $d_v \in \mathbb{Z}$ for all v , and in particular

$$d_1 = \sum_{t \in (\mathbb{Z}/m\mathbb{Z})^*} c_t \{t/m\} = \sum_{\substack{0 \leq t \leq m-1 \\ \gcd(t, m) = 1}} c_t t/m \in \mathbb{Z}.$$

Thus

$$\delta = \gamma\Theta = \sum_{t \in (\mathbb{Z}/m\mathbb{Z})^*} c_t (\sigma_t - t)\Theta + \sum_{\substack{0 \leq t \leq m-1 \\ \gcd(t, m) = 1}} c_t t\Theta = \sum_{t \in (\mathbb{Z}/m\mathbb{Z})^*} c_t \Theta_t + md_1\Theta,$$

and the lemma follows since $c_t \in \mathbb{Z}$, $d_1 \in \mathbb{Z}$, and $m\Theta = (m+1 - \sigma_{m+1})\Theta = -\Theta_{m+1}$ (note that $\sigma_{m+1} = \sigma_1 = 1$). $\square$

Lemma 3.6.18. *Recall that $d = (q-1)/m$. For all b coprime to m we have $\tau(\omega^{-d})^{\sigma_b - b} \in K_m$.*

Proof. Since m is coprime to p , $\text{Gal}(L_m/K_m)$ is the cyclic group of order $p-1$ formed by the maps α_k such that $\alpha_k(\zeta_p) = \zeta_p^k$ and $\alpha_k(\zeta_m) = \zeta_m$ for $k \in (\mathbb{Z}/p\mathbb{Z})^{*1}$. By Galois theory, to prove the lemma we must show that the left-hand side is invariant by each α_k . Since ω^{-d} has order m and the implicit additive character has order p , we have by Definition 2.5.7 and Lemma 2.5.6,

$$\alpha_k(\tau(\omega^{-d})) = \tau(\omega^{-d}, \psi_k) = \omega(k)^d \tau(\omega^{-d}),$$

hence since α_k and σ_b commute,

¹ It is not possible to use here the notation σ_k since this is reserved for elements of $\text{Gal}(L/K)$, hence those that leave ζ_p invariant.

$$\alpha_k(\tau(\omega^{-d})^{\sigma_b-b}) = \omega(k)^{d(b-\sigma_b)}\tau(\omega^{-d})^{\sigma_b-b},$$

and the lemma follows since $\omega(k)^d$ is a power of ζ_m , whence $\omega(k)^{d\sigma_b} = \omega(k)^{db}$. $\square$

It is now easy to deduce the main theorem on the Stickelberger ideal.

Theorem 3.6.19 (Stickelberger). *The Stickelberger ideal $I_s(m)$ annihilates the class group of $K_m = \mathbb{Q}(\zeta_m)$; in other words, for any $\gamma \in I_s(m)$ and any fractional ideal $\mathfrak{a}$ of K_m the ideal $\mathfrak{a}^\gamma$ is a principal ideal.*

Proof. As in the proof of Corollary 3.6.14, it is sufficient to prove that $\mathfrak{p}_m^\gamma$ is a principal ideal for any $\gamma \in I_s(m)$ and any prime ideal $\mathfrak{p}_m$ coprime to m . Raising the equality of Proposition 3.6.13 to the power $\sigma_b - b$ for b prime to m we obtain

$$\mathfrak{p}_m^{m\Theta_b} = \tau(\omega^{-d})^{m(\sigma_b-b)}\mathbb{Z}_{K_m} = \alpha^m\mathbb{Z}_{K_m},$$

where $\alpha = \tau(\omega^{-d})^{\sigma_b-b} \in K_m$ by the above lemma. Since $\mathfrak{p}_m^{\Theta_b}$ and $\alpha\mathbb{Z}_{K_m}$ are ideals of K_m whose m th powers are equal, by uniqueness of the prime ideal decomposition in the Dedekind domain $\mathbb{Z}_{K_m}$ we deduce that they are equal, and in particular that $\mathfrak{p}_m^{\Theta_b}$ is a principal ideal. Since by Lemma 3.6.17 the Θ_b generate $I_s(m)$, it follows that $\mathfrak{p}_m^\gamma$ is a principal ideal for all $\gamma \in I_s(m)$, as was to be proved. $\square$

It is important to note that the above theorem does not bring any new information on the “plus” part of K_m , but only on the “minus” part. More precisely, we have the following:

Proposition 3.6.20. *Let $m \not\equiv 2 \pmod{4}$ and denote by $K_m^+ = \mathbb{Q}(\zeta_m + \zeta_m^{-1})$ the maximal totally real subfield of K_m . Then for any ideal $\mathfrak{a}$ of K_m^+ and any b coprime to m we have*

$$(\mathfrak{a}\mathbb{Z}_{K_m})^{\Theta_b} = \mathcal{N}_{K_m^+/\mathbb{Q}}(\mathfrak{a})^{b-1}\mathbb{Z}_{K_m}.$$

Proof. Recall that we denote complex conjugation by ι , and that $\iota\sigma_t = \sigma_{m-t}$, so that when $\mathfrak{a} \subset K_m^+$ we have $\sigma_{m-t}(\mathfrak{a}\mathbb{Z}_{K_m}) = \sigma_t(\mathfrak{a}\mathbb{Z}_{K_m})$. Thus

$$(\mathfrak{a}\mathbb{Z}_{K_m})^{\Theta_b} = \prod_{1 \leq t \leq m/2, \gcd(t,m)=1} \sigma_t(\mathfrak{a}\mathbb{Z}_{K_m})^{n_t}$$

with $n_t = \lfloor bt/m \rfloor + \lfloor b(m-t)/m \rfloor = b-1$ as is easily seen, since b and t are coprime to m . Since the σ_t for $1 \leq t \leq m/2$ with $\gcd(t,m)=1$ restrict to $\text{Gal}(K_m^+/\mathbb{Q})$ and since $\mathcal{N}_{K_m^+/\mathbb{Q}}(\mathfrak{a})\mathbb{Z}_{K_m} = \prod_{\sigma \in \text{Gal}(K_m^+/\mathbb{Q})} \sigma(\mathfrak{a})$, the result follows. $\square$

Since $I_s(m)$ is generated by the Θ_b as a $\mathbb{Z}$ -module, it follows from the above proposition that for any ideal $\mathfrak{a}$ of K_m^+ and any $\gamma \in I_s(m)$ the ideal

$(\mathfrak{a}\mathbb{Z}_{K_m})^\gamma$ is the principal ideal generated by a power of the norm of $\mathfrak{a}$, so that Stickelberger's ideal theorem does not bring any new information on the class group of K_m^+ . We will see in Chapter 16 devoted to Catalan's conjecture that there is a beautiful theorem of F. Thaine that partially replaces Stickelberger's theorem for K_m^+ .

3.6.5 Diagonalization of the Stickelberger Element

In this section, we will use some simple analytic results that will be proved in Chapters 9 and 10, but which can easily be proved directly except for the nonvanishing of $L(\chi, 1)$. Thus, in contrast to almost all places in this book, there are a couple of forward references.

Keep the above notation, and in particular $G = \text{Gal}(K_m/\mathbb{Q}) \simeq (\mathbb{Z}/m\mathbb{Z})^*$. Denote as usual by $\widehat{G}$ the group of characters of G , which is noncanonically isomorphic to G by Proposition 2.1.16. For $\chi \in \widehat{G}$ we define

$$e_\chi = \frac{1}{\phi(m)} \sum_{\sigma \in G} \chi(\sigma) \sigma^{-1} \in \mathbb{C}[G].$$

It is immediately checked that the e_χ form a complete set of orthogonal idempotents, in other words that $e_\chi^2 = e_\chi$, that $e_{\chi_1} e_{\chi_2} = 0$ if $\chi_1 \neq \chi_2$, and that $\sum_{\chi \in \widehat{G}} e_\chi = 1$. Indeed, these properties are immediate consequences of the orthogonality of characters (and in fact are equivalent to them). As an immediate consequence we deduce that

$$\mathbb{C}[G] = \bigoplus_{\chi \in \widehat{G}} e_\chi \mathbb{C}[G].$$

Since $e_\chi \neq 0$ all the $e_\chi \mathbb{C}[G]$ are nonzero, and on the other hand the number of terms in the direct sum is equal to $|\widehat{G}| = |G| = \dim_{\mathbb{C}} \mathbb{C}[G]$. It follows that all the terms are 1-dimensional, hence that $e_\chi \mathbb{C}[G] = \mathbb{C}e_\chi$, so that the e_χ form a $\mathbb{C}$ -basis of $\mathbb{C}[G]$. Since $e_\chi \mathbb{C}[G]$ is the principal ideal of $\mathbb{C}[G]$ generated by e_χ , it follows that $\mathbb{C}e_\chi$ is an ideal of $\mathbb{C}[G]$.

Lemma 3.6.21. *Denote by j the canonical isomorphism from $(\mathbb{Z}/m\mathbb{Z})^*$ to G such that $j(t) = \sigma_t$, and complex conjugation σ_{-1} by ι . We have $\Theta e_\chi = \lambda_\chi e_\chi$ with*

$$\lambda_\chi = \begin{cases} \phi(m)/2 & \text{if } \chi \text{ is the trivial character,} \\ 0 & \text{if } \chi(\iota) = 1 \text{ and } \chi \text{ is nontrivial,} \\ -L(\overline{\chi \circ j}, 0) & \text{if } \chi(\iota) = -1. \end{cases}$$

Proof. By definition of e_χ we have

$$\sigma_t e_\chi = \frac{1}{\phi(m)} \sum_{\sigma \in G} \chi(\sigma) \sigma_t \sigma^{-1} = \frac{1}{\phi(m)} \sum_{\sigma \in G} \chi(\sigma \sigma_t) \sigma^{-1} = \chi(\sigma_t) e_\chi.$$